

DIAGNOSTIC PERFORMANCE OF AUTOMATED BREAST ULTRASOUND VERSUS HAND-HELD ULTRASOUND IN DETECTION AND CHARACTERIZATION OF BREAST LESIONS: A SYSTEMATIC REVIEW

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BACKGROUND

- Breast cancer is the most common malignancy in women: 2.3 million new cases and 666,000 deaths globally in 2022.
- HHUS is widely used but limited by operator dependence and poor standardisation.
- ABUS offers reproducible, operator-independent whole-breast volumetric imaging, yet its comparative accuracy versus HHUS remains inconsistently reported.

OBJECTIVES

- Primary : to evaluate the diagnostic performance of ABUS compared with HHUS in detecting and characterising breast lesions.
- Secondary : to describe sources of heterogeneity across comparative studies.
- Outcomes: sensitivity and specificity.

METHODS

- Systematic review with narrative synthesis; PubMed, Google Scholar, Semantic Scholar Included: ABUS vs HHUS studies reporting sensitivity AND specificity.
- Excluded: single-modality studies, no extractable data, reviews, case series <10, duplicate populations.
- Reference standard: histopathology and/or ≥6–12-month follow-up.
- Narrative synthesis was chosen over meta-analysis due to substantial clinical and methodological heterogeneity.
- Risk of bias was assessed using QUADAS-2.

RESULTS

Ten studies (2013–2025) from 5 countries; sample sizes 20–10,381 women. Designs: 6 prospective, 2 retrospective, 1 cross-sectional, 1 randomised trial. QUADAS-2 assessment reveals a predominantly low risk of bias in patient selection and reference standards across most studies.

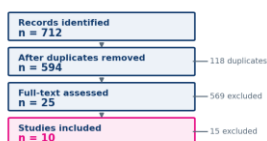


Figure 1. PRISMA flow diagram of study identification and selection.

	ABUS	HHUS
Sensitivity	60.0-95.7	51.1-96.3
Specificity	70.0-97.8	49.4-96.7
AUC	0.79-0.93	0.72-0.93

Table 1. Diagnostic performance ABUS vs HHUS.

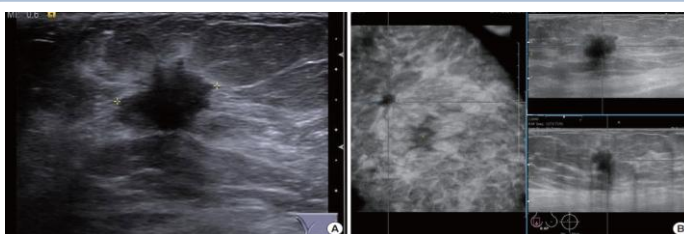


Figure 2. Comparison of a BI-RADS 4B lesion on (A) HHUS and (B) ABUS, histologically confirmed as invasive ductal carcinoma (Reproduced from Kim et al., 2013).

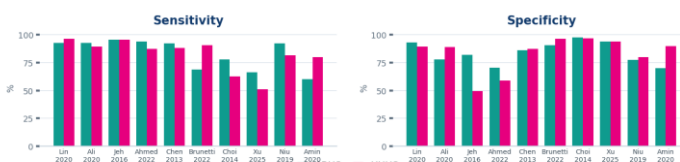


Figure 3. Diagnostic performance per study comparison ABUS vs HHUS.

Key numerical findings

- ABUS was numerically higher for sensitivity in 6/10 studies, whereas HHUS was numerically higher for specificity in 6/10 studies.
- Neither modality was consistently superior across all parameters, performance depended strongly on population, reference standard and operator experience.

CONCLUSION

1. Main finding

ABUS is broadly comparable to HHUS; the direction of advantage varied by study, with no consistent superiority in either sensitivity or specificity.

2. Clinical meaning

ABUS is a credible complementary or alternative modality where operator-independent, reproducible whole-breast imaging is needed.

3. Next step

Paired-design studies with matched operator experience are needed before superiority claims can be made.

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